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Modelling economic policy issues

What is the exchange rate volatility response to COVID-19 and government interventions?

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ABSTRACT

The rapid spread of COVID-19 in 2020 has brought a profound impact on the global economy and forced countries around the world to adopt different intervention measures. Has COVID-19 and these government interventions affected exchange rate volatility? To answer the question, this research explores the impact of COVID-19 and the relevant government response policies on exchange rate volatility in 20 countries during the period of January 13, 2020 to July 21, 2020 by using system GMM estimation. The empirical results indicate that an increase in confirmed cases does significantly raise exchange rate volatility. The various policies adopted by governments in response to the pandemic, such as closing schools, restrictions on internal movements, and public information campaigns also inhibit exchange rate volatility. Furthermore, the economic response policies implemented by governments during the pandemic, including income support, fiscal measures, and international aid, have a restraining effect on exchange rate volatility. Our findings herein provide valuable information and implications for policymakers and financial investors around the world.

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1. Introduction

The outbreak of coronavirus disease 2019 (COVID-19) has caused massive damage to the world economy and severely affected various sectors of society, such as aviation, manufacturing, tourism, and education (Fu and Shen, 2020; Nicola et al., 2020). The International Labor Organization (ILO) estimated 309 million unemployed people in the second quarter of 2020 due to COVID-19, and many enterprises have also been affected by governments' prevention policies, such as lockdowns (Ceylan et al., 2020). It is now forecasted that global GDP in 2020 will fall by 6.7% year-on-year (Apergis and Apergis, 2020; McKibbin and Fernando, 2020). With the virus bringing about unpredictable economic losses, it has become a catalyst for the current global economic crisis.

With the continuous spread of COVID-19, many scholars have tried to explore its impact on the macroeconomy, including industrial production (Altig et al., 2020; Appiah-Otoo, 2020), GDP growth (Vidya and Prabheesh, 2020), household consumption (Binder, 2020; Liu et al., 2020b), employment (Montenovo et al., 2020; Yu et al., 2020), global supply chains (Bonadio et al., 2020; Qin et al., 2020a) and innovation ability (Han and Qian, 2020). However, as an exogenous shock, the impact of COVID-19 on the macroeconomy will take some time to be observed, because there is a certain lag effect, but at the same time, the daily or weekly changes of the pandemic will have a continual influence

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on financial markets (Lan et al., 2020). Most current research on the impact of COVID-19 on financial markets focuses on stock returns (Iyke, 2020c; Liu et al., 2020a; Prabheesh, 2020; Salisu and Sikiru, 2020; Yan and Qian, 2020) and exchange rate (Iyke, 2020b; Narayan, 2020b; Narayan et al., 2020a). Other scholars are concerned about the impact of COVID-19 on oil prices (Devpura, 2020; Devpura and Narayan, 2020; Gil-Alana and Monge, 2020; Iyke, 2020a; Prabheesh et al., 2020).

Social distancing and lockdowns caused by COVID-19 have also had profound impacts on society. Some studies have shown that COVID-19 has certain effects on the labor market (Béland et al., 2020; Coibion et al., 2020; Kahn et al., 2020), and that factors such as education level (Bick et al., 2020), age (Adams-Prassl et al., 2020), and immigration (Borjas and Cassidy, 2020) play different regulatory roles. On the other hand, results exist showing that COVID-19 does not impact physical health (Goldstein and Lee, 2020), but instead harms a person's mental health (Lu et al., 2020) and well-being (Brodeur et al., 2020). Some studies have presented that COVID-19 increases gender (Alon et al., 2020) and racial inequalities (Borjas and Cassidy, 2020), while others noted that COVID-19 may have a certain positive impact on the environment (Cicala et al., 2020) and air quality (Ming et al., 2020).

The literature has also explored the influence of government intervention policies on economic activities (Ertuğrul et al., 2020; Haldar and Sethi, 2020; Phan and Narayan, 2020; Song et al., 2020). Compared to SARS, Ebola, and MERS, government restrictions on individual mobility and voluntary social distancing during COVID-19 have caused greater stock market volatility (Baker et al., 2020; Mishra et al., 2020; Narayan, 2020a). However, with the implementation of government policy interventions, market liquidity can be effectively improved (Haroon and Rizvi, 2020; He et al., 2020; Narayan et al., 2020b; Sha and Sharma, 2020; Sharma, 2020).

The traditional finance literature has pointed out that the magnitude of exchange rate volatility has great significance over the stability of a country's foreign trade and external environment. Higher exchange rate volatility intensifies financial market risks, increases foreign investment uncertainty, and leads to a reduction of social welfare (Byrne and Davis, 2005; Devereux, 2004). Research in recent years has found that unforeseen information plays a unique role in predicting exchange rate volatility. Narayan et al. (2018) presented that terrorist attacks have a significant impact on the bilateral exchange rates of various countries. After terrorist attacks, some national currencies have appreciated, while some national currencies have depreciated and are continuously affected. Similarly, Sharma et al. (2019) explored the relationship between government shutdowns and exchange rate fluctuations, showing results that exchange rate volatility tends to increase due to shutdowns.

With the global outbreak of COVID-19, various countries have successively adopted measures such as transportation control and population movement restrictions, which have made the global economic and trade environment even harsher and brought further impacts to international trade. According to statistics from the World Trade Organization (WTO), the volume of merchandise trade in the first quarter of 2020 fell by 3% year-on-year.[1] Preliminary estimates of global trade in the second quarter of 2020 show that the epidemic and related blockade measures have affected a large part of the world's population, with global merchandise trade dropping even more year-on-year by 18.5%. Due to the sharp decrease in global trade volumes, it is clearly foreseeable that international capital flows will inevitably be affected (Qin et al., 2020b), which will significantly increase exchange rate volatility. With the reduction of global capital flows, financial institutions are assuming an upward trend in the risks caused by the unbalanced demand for international financial assets, causing investors to change their compensation for currency holdings and thereby impacting exchange rate volatility (Debelle, 2020; Gabaix and Maggiori, 2015).

The above-mentioned research has offered some enlightenment and tested whether COVID-19 and government intervention affect exchange rate fluctuations. Fig. 1 illustrates the exchange rate changes of four currencies which are EUR, RMB, GBR, YEN from January 1, 2019 to July 31, 2020. As seen there, following the outbreak of COVID-19 in January 2020, the exchange rates of the four currencies have experienced varying degrees of fluctuations. We thus calculate the standard deviations of the four currencies before and after January 2020 to observe exchange rate volatility.[2] The results show that, with the exception of RMB, the standard deviations of the other three currencies during COVID-19 are greater than the same period of the previous year 2019, indicating that the three major currencies did experience greater fluctuations during the pandemic (Chang et al., 2021). The reason for the small standard deviation in RMB during the pandemic may be due to China's strong early control measures, which prevented the spread of the pandemic in a timely manner and thus did not cause drastic fluctuations.

This research thus applies system GMM (SGMM) estimation to explore the impacts of COVID-19 and related government responses on exchange rate volatility by using panel data from 20 countries spanning January 13 to July 21, 2020. The paper selects the logarithm of biweekly confirmed cases as an indicator to measure the development or slowdown of COVID-19. Since the prevention and control measures adopted by each country and the timeliness of the implementation of the measures are different, the research chooses the Oxford COVID-19 Government Response Tracker (OxCGRT) as the main research object, which is built to provide a comprehensive measurement methodology to understand how a government's response has changed throughout the period of COVID-19 (Hale, 2020). From theoretical and empirical tests, this paper illustrates the impacts of COVID-19 and government intervention on exchange rate volatility in order to help fill the gaps in existing literature.

[1] The data come from https://unctad.org/en/Pages/Home.aspx.

[2] This is available in the Appendix.

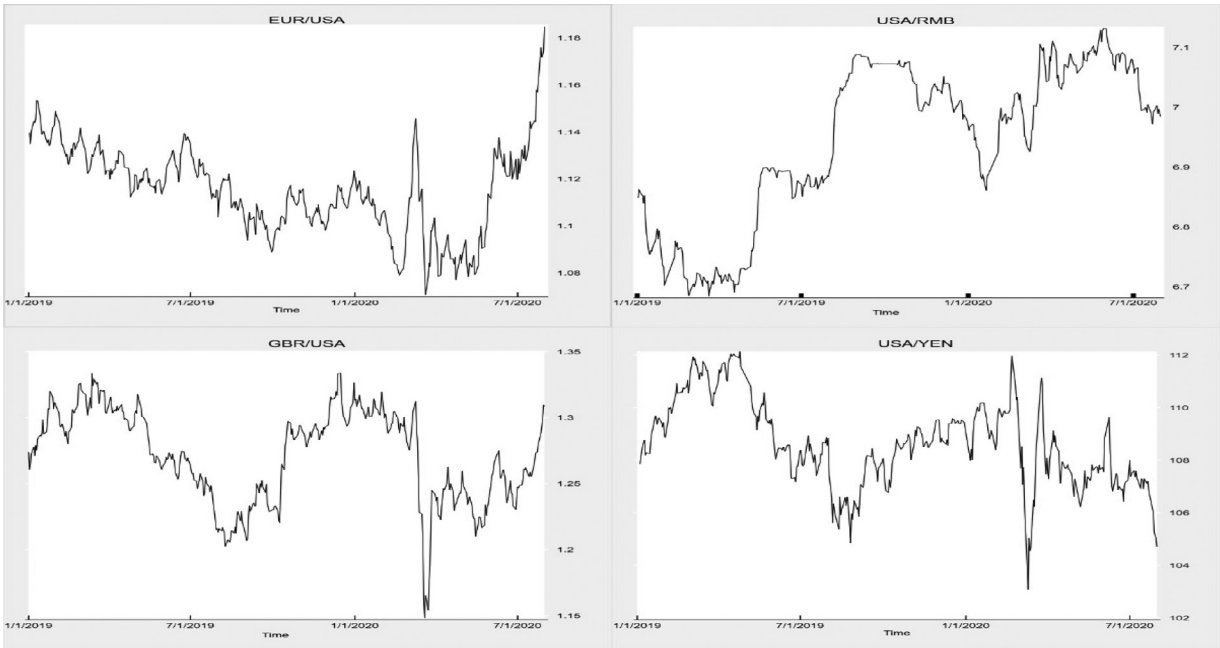

Fig. 1. Changes in the exchange rate of the four currencies from January 1, 2019 to July 31, 2020.
Note: Plot of the exchange rate changes of four currencies (EUR, RMB, GBR, YEN) before and after the pandemic. The data sample is 01/1/2019 to 31/7/2020.

The empirical results of this paper present that an increase in the percentage of biweekly confirmed cases indeed intensifies exchange rate volatility. However, the different intervention measures taken by governments have suppressed exchange rate volatility. Specifically, non-pharmaceutical government interventions such as closing schools, restricting domestic travel, and public information campaign policies can curb exchange rate volatility. Similarly, economic support policies , such as income support, fiscal stimulus, and international aid, all help curb exchange rate fluctuations. In order to ensure the reliability of the conclusions herein, in the robustness test we change the variable of confirmed cases every two weeks to confirmed death every two weeks and perform sub-sample regression. Most regression results support the original conclusions of this article.

Through the discussion of the above-mentioned related literature, one may see that although many scholars discussed the impact of COVID-19 on the macroeconomy, financial markets, and society, scant studies have attempted to illustrate whether COVID-19 causes exchange rate fluctuations and the impacts of various government response measures on exchange rate volatility. Thus, our research mainly fills the gap in the existing literature in the following aspects. First, our findings discuss for the first time the relationship between COVID-19 and exchange rate volatility. While most studies have emphasized the impact of COVID-19 on the global economy and stock markets (Sharif et al., 2020; Zhang et al., 2020), we turn attention towards exchange rate volatility and look to provide a more complete analysis of the link between COVID-19 and exchange rate volatility. Second, we consider the impacts of government intervention measures adopted to prevent the spread of the pandemic on exchange rate fluctuations. Research has noted that strong control measures implemented by governments can effectively curb the spread of COVID-19 and decrease exchange rate volatility. The various fiscal support policies implemented by governments around the world have also played a positive role in curbing exchange rate fluctuations. Through the above contributions, we offer concrete recommendations for the implementation of government policies in the future.

The rest of the paper runs as follows. Section 2 introduces the methodology employed in the empirical analysis. Section 3 is the data description and source. Section 4 presents the empirical results and the robustness test. Section 5 is the conclusion.

2. Methodology

Following the classical literature on exchange rate volatility and pandemic (Gabaix and Farhi, 2015; Kočenda and Moravcová, 2019; Noy and Vu, 2010; Oseni, 2016), this paper uses panel data from 20 countries for empirical analysis. Compared with cross-sectional data, panel data take multiple observations to facilitate the inference of causality and also to incorporate the hysteresis of behavioral decision-making into a model for discussion (Wooldridge, 2016). In order to address the potential endogeneity problem, this paper adopts the system GMM regression, which is:

$$ER_{i,t} = \alpha ER_{i,t-1} + \beta_1 Log(Bcase_{i,t}) + \beta_2 X_{it} + \beta' X + \mu_i + \eta_t + \varepsilon_{it}, \quad (1)$$

For the dependent variable $ER_{i,t}$, we follow Devpura and Narayan (2020) and use log percentage returns to measure exchange rate volatility, such that an increase denotes a depreciation of the quoting currency; $ER_{i,t-1}$ represents the dependent variable lags by one period; $Log(Bcase_{i,t})$ represents the logarithm of biweekly confirmed cases in 20 sample countries; X_{it} is equivalent to a set of explanatory variables representing the degree of government intervention; X corresponds to a set of other explanatory variables; μ_i is the country-level fixed effect; η_t is the time-specific effect; ε_{it} is an error term; and i and t indicate countries and periods, respectively.

We estimate that Eq. (1), as a typical dynamic equation, can solve the problems of endogeneity of equations and missing dependent variables through the generalized moment model (GMM) model. In general, GMM regression estimation contains two types of methods: GMM difference estimation and GMM system estimation.

Under different GMMs, because the autoregressive process becomes too persistent, or the ratio of the variance of the panel layer effect to the variance of the specific error is too large, the lag-level tools in the Arellano–Bond estimator become weak (Arellano and Bond, 1991). Therefore, Blundell and Bond (1998) proposed a system estimator that employs moment conditions in which lagged differences are used as instruments for the level equation in addition to the moment conditions of lagged levels as instruments for the difference equation. Compared with different GMM, the advantage of system GMM (SGMM) is that it improves the efficiency of estimation and can estimate the coefficients of variables that do not change with time. This paper thus chooses the system GMM as the regression technique.

3. Data description and source

We first collect the bilateral exchange rate data of 20 countries from January 13 to July 21, 2020 through the Wind database.[3] Next, we calculate the bilateral exchange rate of each country and convert it into daily percentage log returns ($ER_{i,t}$) defined as $ER_{i,t} = \ln(S_{i,t}/S_{i,t-1}) * 100\%$, where $S_{i,t}$ is the daily bilateral exchange rate at time t. A negative change indicates an appreciation of the selected currency. Taking the Japanese yen as an example, when $ER_{i,t}$ decreases, it means that the amount of yen needed to buy one unit of US dollar has decreased, indicating that the yen has appreciated against the US dollar.

We then collect data on biweekly confirmed cases and biweekly confirmed deaths for 20 sample countries from https://ourworldindata.org/. The numbers of most confirmed cases and deaths have increased exponentially, the number of confirmed cases in some countries within a period of time is 0, and the total population of each country is different. Therefore, this paper first adds one to the data where the number of confirmed and dead cases is zero and then takes the logarithm of biweekly confirmed cases and confirmed deaths in the 20 sample countries as an explanatory variable, which also contributes to eliminating heteroscedasticity and avoiding spurious regression (Wooldridge, 2016). Fig. 2 shows the growth trend of confirmed cases and deaths in the sample countries.

In order to explore the relationship between the degree of government response and exchange rate volatility, we collect the classification indicators and comprehensive indicators of 20 sample countries through the Oxford COVID-19 Government Responses Tracker (*OxCGRT*), which come from https://ourworldindata.org/. *OxCGRT* summarizes the epidemic prevention measures taken by governments and divides them into three dimensions: containment and closure, economic response, and health systems.[4] Containment and closure include school closings (*C1*), workplace closings (*C2*), canceling public events (*C3*), restrictions on gathering size (*C4*), public transport closure (*C5*), stay at home requirements (*C6*), restrictions on internal movement (*C7*), and restrictions on international travel (*C8*). The economic response category contains four indicators: income support (*E1*), debt/contract relief for households (*E2*), fiscal measures (*E3*), and giving international support (*E4*). The last item is health systems, covering public information campaign (*H1*), testing policy (*H2*), contact tracing (*H3*), emergency investment in health care (*H4*), and investment in COVID-19 vaccines (*H5*). The above indicators are of two types: ordinal and numeric.[5]

For comprehensive indicators including stringency index (*ST*), containment and health index (*CH*), economic support index (*ES*), and overall government response index (*GR*), all of the indices are simple averages of the individual component indicators.[6] For example, *ST* includes a total of nine response indicators, containing all indicators under the Containment and closure and the *H1* indicator, which belongs to the Health systems category. The score ranges from 0 to 100, and the higher the score is, the more stringent the government response is to the COVID-19 pandemic. Fig. 3 shows the change in the stringency index for the sample countries from January 13 to July 21, 2020. We see from the above explanations of various indicators that the measurement of government intervention herein covers various specific measures and also helps discuss the impact of government intervention on exchange rate volatility more deeply through four comprehensive indicators that focus on different aspects.

[3] The 20 sample countries include Australia, Brazil, Canada, Switzerland, China, Germany, Spain, Finland, United Kingdom, India, Italy, Japan, South Korea, Norway, New Zealand, Portugal, Russia, Singapore, Sweden, and the U.S. Among the 20 countries, the U.S. selects a bilateral exchange rate of US dollar to pound sterling, and the remaining 19 countries use their local currency's units per US$.

[4] See the Appendix for specific classification and composition.

[5] Among the 17 specific indicators, *E3*, *E4*, *H4*, and *H5* are measured by number — typically the value is in USD. In the empirical regression, we add one to the values of these indicators and take the logarithmic value as the explanatory variable. The remaining indicators measure policies on a simple scale of severity/intensity.

[6] The equation is $index = \frac{1}{k}\sum_{j=1}^{k} I_j$, where k is the number of component indicators in an index, and I_j is the sub-index score for an individual indicator. See the Appendix for details.

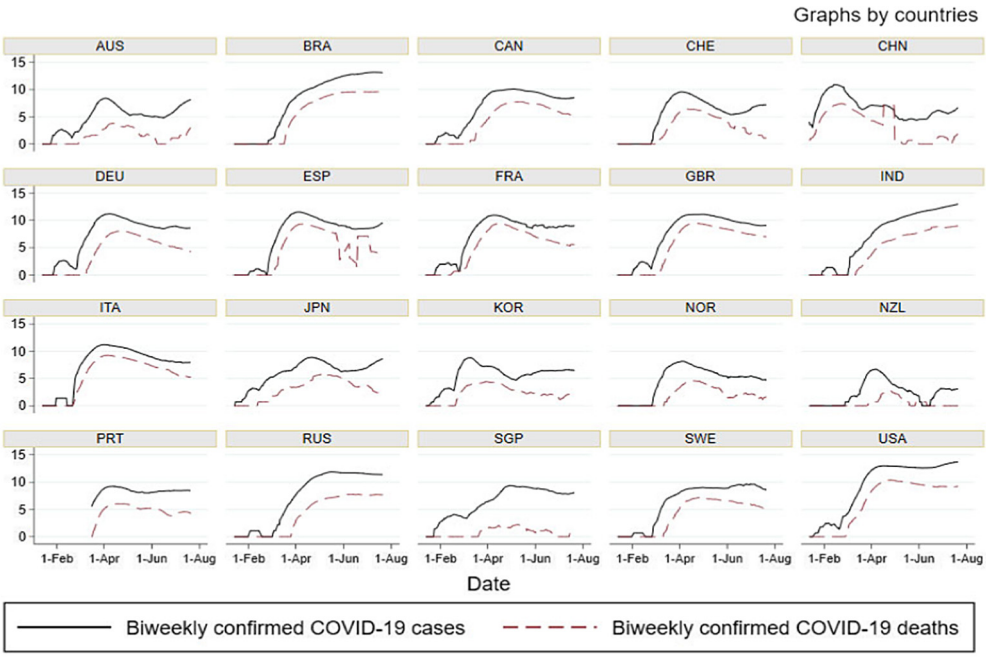

Fig. 2. Biweekly confirmed COVID-19 cases and deaths in the 20 countries.
Notes: Fig. 2 shows the growth trend of the number of confirmed cases and deaths in 20 countries. The 20 sample countries and corresponding country codes are Australia (AUS), Brazil (BRA), Canada (CAN), Switzerland (CHE), China (CHN), Germany (DEU), Spain (ESP), Finland (FRA), United Kingdom (GBR), India (IND), Italy (ITA), Japan (JPN), South Korea (KOR), Norway (NOR), New Zealand (NZL), Portugal (PRT), Russia (RUS), Singapore (SGP), Sweden (SWE), and the United States (USA). The data sample is 13/1/2020 to 21/7/2020.

For the control variables, this paper selects the interest rate (*Interest Rate*), the logarithm of foreign exchange reserves (*Log(FER)*), and the logarithm of the consumer price index (*Log(CPI)*) of each country. The control variables selected above have an impact on a country's currency market following Andersen et al. (2003) and Fatum et al. (2012).

Table 1 presents the statistical properties of the variables used in the study: logarithms of biweekly confirmed COVID-19 cases (*Log(Bcases)*) and the exchange rate volatility (*ER*). The remaining variables correspond to the four comprehensive indicators and seventeen specific measures of government response and control variables. From the table we see for all sample countries that the mean value of volatility is 0.038, the standard deviation is 6.221, and the maximum is 97.462, implying there exist big differences in exchange rate volatility among the sample countries. For the comprehensive indicators, the mean of *ES* is 44.953 and the standard deviation is 35.063, denoting that there exist great differences in economic compensation that the governments have invested in the pandemic, which may be caused by the disparity in the size of the economy and the severity of the pandemic of each country. By comparing the mean values of *ST*, *GR*, *CH* and *EC*, we find that the mean value of *CH* is greater than the other three indicators, implying that the government response policies implemented by various countries are mostly focused on containment and closure measures and maintenance of the health systems.

We now observe the statistical properties of 17 specific measures and show that the mean value of *C8* (2.445) is the highest, indicating that the sample countries first chose to restrict international travel when implementing containment and closure measures, while the mean value of *C7* is only 0.973, illustrating that sample countries have relatively loose restrictions on the movement of people within their own countries. As for *Log(E3)*, the mean of it is 0.861, with a standard deviation of 4.26, implying that the fiscal measures implemented by governments in the face of the pandemic are quite different. Regarding the control variables, the lowest value of interest rates in various countries is −0.97% and the highest value is 6.37%, whereas their standard deviation is 1.742, meaning there exists a big difference among various countries. As for *Log(FER)*, the minimum, maximum, mean, and standard deviation are respectively 8.664, 14.952, 11.926, and 1.464, denoting that there is a certain degree of difference in the balance of foreign exchange reserves held by various governments. The mean of *Log(CPI)* is 4.586, with a standard deviation of 0.086, indicating that there is no significant difference in the consumer price index in the sample countries.

4. Empirical results and robustness test

4.1. Empirical results

(1) Empirical results for Log(Bcases) and comprehensive indicators

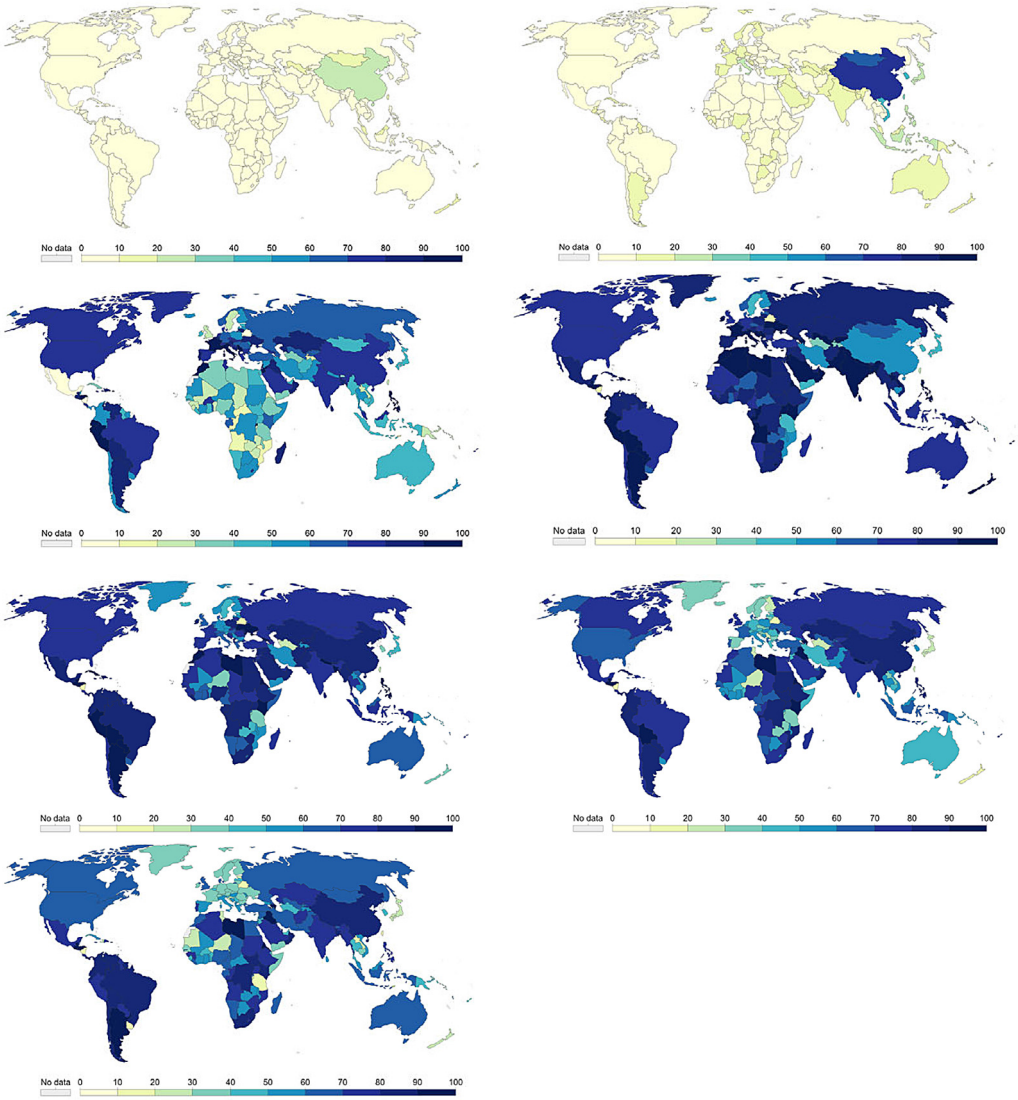

Fig. 3. Worldwide trend of GRSI.
Notes: Fig. 3 shows the trend of the global government stringency index, whose score ranges from 0 to 100. The higher the score is, the more stringent the government response is to the COVID-19 pandemic. The data sample is 13/1/2020 to 21/7/2020. The source of Fig. 3 is the website https://ourworldindata.org/grapher/cOVID-stringency-index.

We now observe whether the development of COVID-19 and government responses to it exert influences on exchange rate volatility. Table 2 presents the results of the system GMM estimates of *Log(Bcases)* and four comprehensive indicators. Column 1–4 presents *ST*, *GR*, *CH*, and *ES*.

The premise of using the system GMM for estimation is that there is no autocorrelation in the disturbance term ε_{it}. The Arellano and Bond test of the second-order autocorrelation does not reject the null hypothesis in all equations, revealing that the results from SGMM estimates are consistent. As an over-identification test, Sargan's test results do not reject the null hypothesis, which proves the validity of instrumental variables.

The parameter estimation value in Table 2 provides evidence that the increase in the percentage of biweekly confirmed cases has indeed boosted exchange rate volatility at a 5% significance level. For example, in column (1) the regression coefficient of biweekly confirmed cases is 0.664, which means that when biweekly confirmed cases grow by 1%, the exchange rate return increases by 0.664%. The regression results in columns (1)–(4) all confirm that the uncertainty caused by COVID-19 has indeed significantly increased exchange rate volatility. Our finding is similar to that of Sharif et al. (2020), who found that the rise in the number of confirmed cases increases volatility in the U.S. financial market. As Obstfeld et al. (1996) claimed, a high level of economic policy uncertainty causes investors to adjust their expectations for policies and the economy, leading to exchange rate volatility.

Table 1
Summary of descriptive statistics.

Variable	Obs	Mean	Std.Dev.	Min	Max
Log(Bcases)	2597	6.539	3.837	0	13.701
ER	2640	.038	6.221	−97.616	97.462
ST	2575	48.555	29.88	0	100
GR	2575	49.986	27.319	0	96.15
CH	2579	50.861	27.648	0	100
ES	2587	44.953	35.063	0	100
C1	2575	1.735	1.361	0	3
C2	2575	1.357	1.187	0	3
C3	2575	1.284	.894	0	2
C4	2575	2.247	1.809	0	4
C5	2575	.475	.685	0	2
C6	2575	.905	.995	0	3
C7	2577	.973	.895	0	2
C8	2582	2.445	1.442	0	4
E1	2588	.955	.849	0	2
E2	2588	.957	.839	0	2
Log(E3)	2578	.861	4.26	0	28.303
Log(E4)	2586	.172	1.801	0	27.45
H1	2586	1.713	.682	0	2
H2	2589	1.695	.976	0	3
H3	2586	1.244	.744	0	2
Log(H4)	2585	.528	2.993	0	26.214
Log(H5)	2583	.254	2.042	0	21.193
Interest Rate	2640	.942	1.742	−.97	6.37
Log(FER)	2640	11.926	1.464	8.664	14.952
Log(CPI)	2640	4.586	.086	4.281	4.743

Table 2
GMM estimation: Log(Bcases) and four comprehensive indicators.

Variable	(1)	(2)	(3)	(4)
L.Earn	0.072*	0.009	0.051	0.014
	(1.861)	(0.142)	(1.046)	(0.246)
Log(Bcase)	0.664**	1.731**	1.143**	0.669**
	(2.163)	(2.497)	(2.026)	(2.495)
ST	−0.072**			
	(−2.221)			
GR		−0.207**		
		(−2.514)		
CH			−0.137**	
			(−2.047)	
ES				−0.082**
				(−2.510)
Interest Rate	−0.391***	−0.495***	−0.486***	−0.346***
	(−4.436)	(−4.306)	(−4.837)	(−3.854)
Log(FER)	1.298***	1.803***	1.991***	1.005**
	(5.214)	(4.758)	(5.081)	(2.117)
Log(CPI)	3.622	5.421*	4.626*	2.182
	(1.619)	(1.892)	(1.674)	(1.325)
Constant	−32.509***	−46.690***	−44.935***	−22.278**
	(−3.064)	(−3.067)	(−2.860)	(−2.392)
Observations	1,997	1,997	1,997	1,997
AR(1) (*p*-value)	0.001	0.001	0.001	0.001
AR(2) (*p*-value)	0.379	0.106	0.166	0.755
Sargan(*p*-value)	0.168	0.261	0.287	0.172

Notes: *z* statistics are in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. AR(1) and AR(2) denote the p-value for the Arellano–Bond test of first-order and second-order autocorrelations, respectively. Sargan represents the p-value of the over-identification test.

Once the lagged dependent variable and the COVID-19 pandemic variable are controlled, we find that the variable *ST* is negative and statistically significant at the 5% level, illustrating that when government adopts more stringent anti-epidemic measures, it can curb exchange rate volatility. Similarly, the variables of *GR* and *CH* are significantly negative, indicating that the government response index and containment and health index overall are able to suppress exchange rate volatility. On average, the implementation of any physical distance intervention can reduce overall incidences of COVID-19 by 13% (Islam et al., 2020). Therefore, the implementation of powerful government measures sends a strong

signal to the market and investors that the government has the ability to control the development of the epidemic, thus increasing investors' confidence, stabilizing the financial market, and curbing exchange rate volatility (Huang and Zheng, 2020).

For the *ES* index, the regression coefficient also significantly suppresses exchange rate volatility at the 5% level, and the empirical result shows that with the strengthening of a government economic support policy by one index point, the exchange rate return decreases 0.082%. The *ES* index in the Appendix includes specific measures in two aspects: income support and household debt relief. Regarding the former measure, a government's direct fiscal expenditure to provide economic assistance to low-income households can be regarded as an expansionary fiscal policy that increases private sector consumption and prohibits exchange rate volatility (Monacelli and Perotti, 2010). For household debt relief measures, Jongwanich and Kohpaiboon (2013) found that a rapid rebound in capital flows may lead to excessive currency appreciation, especially when capital flows are in the form of bank loans. A government's cancellation of loan repayments and other measures causes dramatic changes in capital flows, which in turn have a negative impact on exchange rate volatility.

For the control variables, the regression results for the interest rate in columns (1) to (4) all show a significantly negative impact on exchange rate fluctuations, while foreign exchange reserves have the opposite effect. For the logarithms of the consumer price index in columns (2) and (3), all have a promoting effect on exchange rate volatility. Goldberg and Campa (2010) analyzed the corresponding consumer price index (CPI) and exchange rate data of 21 industrialized economies and noted that the distinctive channels that constitute CPI have different degrees of sensitivity to exchange rates, revealing a correlation between the consumer price index and exchange rate volatility.

(2) Empirical results for Log(Bcases), containment, and closure

Table 3 presents the regression results of *Log(Bcases)*, containment, and closure sub-index. Columns (1) to (8) correspond to the specific government measures included in the sub-indices. Similar to the regression results in Table 2, the regression results of columns (1), (5), (6), and (7) are respectively significant at the 5% level. This indicates that when the government's response policies change from comprehensive indicators to the containment and closure sub-index, the increase in the percentage of biweekly confirmed cases still lifts exchange rate volatility.

The regression results of governments' responses (*C1-C8*) show different conclusions. Table 3 presents that for the regression coefficients of *C1-C8*, only *C3* is positive, and the other coefficients are all negative, among which *C6* is significant at the 1% level, while *C1* and *C5* only pass the significance test at the 10% level. Among all the regression results, the maximum regression coefficient of *C7* is −3.418, illustrating that when the intensity of the restrictions on internal movement policy increases by one index point, exchange rate volatility decreases by 3.418%. For the regression coefficient of *C1*, the result is −0.747 and only passes the 10% significant test, indicating that for every index point rise in the degree of a strict school closing policy, exchange rate volatility drops by 0.747%. The regression results of *C6* pass the 1% significance test, indicating that stay-at-home requirements significantly suppress exchange rate fluctuations. However, the regression result of *C5* only passes the 10% significance test, proving that the government's closure of public transportation measures do not have a very significant impact on exchange rate fluctuations.

From the overall regression results, the containment and closure measures adopted by the government have the effect of restraining the exchange rate volatility to different degrees. Although some studies believe that the containment and closure measures implemented by governments have a huge impact on the economy, the implementation of various venue closures and travel restrictions can effectively reduce the increase in confirmed cases (Chinazzi et al., 2020; Prem et al., 2020), thereby reducing the increase in economic policy uncertainty, alleviating market panic, and inhibiting exchange rate volatility.

(3) Empirical results for Log(Bcases) and economic response

The economic support index shown in Table 2 is a comprehensive index calculated by adding up the two measures *E1* and *E2*. Table 4 lists the regression results of each specific measure.

Column (1) shows the impact of income support policies on exchange rate volatility. The income compensation policy includes whether the government pays wages or provides direct cash payments for the unemployed or those unable to work, general basic income, or similar expenses. The coefficient of *E1* is −2.679 and is significantly negative at the 10% level. This finding indicates that when the intensity of the income support policy adopted by the government increases by one index point, exchange rate volatility decreases by 2.679%. COVID-19 has caused many companies to stop work and production, workers' wages have decreased, and the income gap has widened. The income subsidy policy implemented by a government can help reduce the income gap, promote consumption, and curb exchange rate fluctuations (Jeanneney and Hua, 2001; Lima and Porcile, 2013).

Columns (3) and (4) respectively show the regression results of E3 and E4, indicating the impact of fiscal policy and international aid on exchange rate fluctuations. The regression results of these two policies have passed the significance test at the 5% level, and the regression coefficients are −0.015 and −0.014, respectively. This indicates that a government's fiscal stimulus policy and international aid can restrain exchange rate fluctuations to a certain extent.

Column (2) shows the effects of the debt/contract relief for households policy on exchange rate volatility. The regression result of this policy is not significant, indicating that a government's reduction of household debt does not have a significant impact on exchange rate volatility.

(4) Empirical results for Log(Bcases) and health system

We now turn our attention to specific measures under the government health system. Table 5 shows the regression results for *Log(Bcases)* and the sub-indices of the health system. Liang et al. (2020) illustrated that information from

Table 3
GMM estimation: Log(Bcases), containment, and closure.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
L.Earn	0.079**	0.126***	0.691*	0.198***	0.097**	0.083**	0.216***	0.115**
	(2.060)	(2.823)	(1.856)	(2.731)	(2.377)	(2.228)	(2.589)	(2.490)
Log(Bcase)	0.426**	0.325	−0.238	0.581	0.365**	0.285**	0.878**	0.158
	(2.029)	(1.328)	(−0.809)	(1.304)	(2.088)	(2.163)	(2.042)	(0.620)
C1	−0.747*							
	(−1.837)							
C2		−1.562						
		(−1.371)						
C3			0.876					
			(0.731)					
C4				−1.109				
				(−1.539)				
C5					−1.090*			
					(−1.849)			
C6						−0.975***		
						(−2.716)		
C7							−3.418**	
							(−1.977)	
C8								−0.115
								(−0.315)
Constant	−116.800***	−20.381	−32.129**	120.948	−13.091	−56.132***	−67.985**	−15.805
	(−4.924)	(−0.306)	(−2.026)	(1.199)	(−0.317)	(−3.179)	(−2.412)	(−0.415)
Observations	1,997	1,997	1,997	1,997	1,997	1,997	1,999	2,002
AR(1) (*p*-value)	0.001	0.001	0.005	0.001	0.001	0.001	0.001	0.001
AR(2) (*p*-value)	0.108	0.122	0.307	0.122	0.485	0.931	0.410	0.294
Sargan(*p*-value)	0.117	0.161	0.163	0.851	0.100	0.122	0.394	0.390

Notes: The same as Table 2. In view of the limited space, the control variable, fixed effect are not reported, but are available upon request.

Table 4
GMM estimation: Log(Bcases) and economic response.

Variable	(1)	(2)	(3)	(4)
L.Earn	0.045	0.075**	0.063	0.089**
	(0.649)	(2.138)	(1.435)	(2.547)
Log(Bcase)	0.426	0.054	0.082	0.003
	(1.371)	(1.049)	(0.559)	(0.190)
E1	−2.679*			
	(−1.768)			
E2		−0.375		
		(−1.375)		
log(E3)			−0.015**	
			(−2.341)	
log(E4)				−0.014**
				(−2.245)
Constant	77.362	−68.269***	19.517	−80.943***
	(0.893)	(−3.775)	(0.308)	(−3.475)
Observations	2,007	2,007	1,998	2,005
AR(1) (*p*-value)	0.001	0.001	0.001	0.001
AR(2) (*p*-value)	0.169	0.773	0.142	0.887
Sargan(*p*-value)	0.340	0.147	0.152	0.238

Notes: *z* statistics are in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. AR(1) and AR(2) denote the p-value for the Arellano–Bond test of first-order and second-order autocorrelations, respectively. Sargan represents the p-value of the over-identification test. In view of the limited space, the control variable, fixed effect, and other results are not reported, but are available upon request.

online news and social media is the main channel for most investors to understand the market dynamics. Therefore, a government's daily positive reports on the degree of control of this pandemic via traditional and social media can reduce investor panic and stabilize the market. Through testing policies and contact tracing measures, a government can effectively slow down the spread of COVID-19 and reduce the number of confirmed cases, while emergency medical input and vaccine research can improve investors' confidence in pandemic prevention and control, thus curbing exchange rate volatility caused by any panic. However, from Table 5 we see that among the various sub-indices under the health system, only the public information campaign policy has a significantly negative correlation effect, and although the regression coefficients of the other indicators are negative, they are insignificant. The regression coefficient of *H1* is −4.062 and

Table 5
GMM estimation: Log(Bcases) and health systems.

Variable	(1)	(2)	(3)	(4)	(5)
L.Earn	0.092*	0.097**	0.063	0.099**	0.098**
	(1.675)	(2.230)	(1.467)	(2.123)	(2.066)
Log(Bcase)	1.266***	0.453	0.151	0.099	0.056
	(2.943)	(1.066)	(0.925)	(0.626)	(0.345)
H1	−4.062***				
	(−2.818)				
H2		−1.300			
		(−1.142)			
H3			−0.443		
			(−1.264)		
log(H4)				−0.001	
				(−0.222)	
log(H5)					−0.003
					(−0.287)
Constant	−88.960	−83.034	−23.583	17.122	22.372
	(−1.551)	(−0.871)	(−0.536)	(0.279)	(0.382)
Observations	2,005	2,007	2,005	2,004	2,004
AR(1) (*p*-value)	0.001	0.001	0.001	0.001	0.001
AR(2) (*p*-value)	0.077	0.264	0.102	0.078	0.080
Sargan(*p*-value)	0.101	0.269	0.595	0.204	0.096

Notes: *z* statistics are in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. AR(1) and AR(2) denote the p-value for the Arellano–Bond test of first-order and second-order autocorrelations, respectively. Sargan represents the p-value of the over-identification test. In view of the limited space, the control variable, fixed effect, and other results are not reported, but are available upon request.

significant at the 1% level, indicating that when a public information campaign is expanded by one index point, exchange rate volatility falls by 4.062%.

4.2. Robustness test

This paper previously examines the impact of COVID-19 and different government interventions on exchange rate volatility and indicates that exchange rate volatility has intensified along with the continuous spread of COVID-19. For government intervention measures, strict closures and a containment and economic support policy can restrain exchange rate volatility to a certain extent. To further confirm the robustness of the regression results, we first replace the main explanatory variable that measures the extent of the pandemic. Second, in view of the rapid spread of the pandemic in China in the early stage, we hereby remove Chinese data and perform a sub-sample regression. It turns out that the results of the robustness test are basically consistent with the empirical results.

(1) Robustness test: Log(Bdeaths) and four comprehensive indicators

For the baseline analysis, we use biweekly confirmed COVID-19 cases to indicate the severity of the COVID-19 pandemic and identify the impact of the COVID-19 outbreak on exchange rate volatility. However, another more important indicator for measuring the severity of a pandemic is mortality. Therefore, this paper selects the logarithm of biweekly confirmed deaths as a substitute variable to conduct a robustness test to check whether our main findings still make sense when the alternative measure is employed.

Table 6 shows the results of the robustness test for the four comprehensive government response indicators. We observe that all results are similar to the baseline results in Table 2. After replacing *Log(Bcases)* with *Log(Bdeaths),* the increase in the percentage of biweekly confirmed deaths also raises exchange rate volatility. It can be seen that even after changing the explanatory variables that measure the extent of COVID-19, the conclusion that the pandemic promotes exchange rate volatility remains unchanged. As for the comprehensive indicators, *ST*, *CH*, and *GR* still produce a significant inhibiting effect on exchange rate volatility. Among them, the government response index has the strongest inhibitory effect, with a correlation coefficient of −0.227, which is significant at the 5% level. For the stringency index *(ST)* and containment and health index *(CH)*, the regression coefficients are respectively −0.114 and −0.153, which are also significant at the 5% level. However, the regression result of the economic support index (*ES*) is not significant, although the regression coefficient is negative. Accordingly, we believe that the impacts of the four comprehensive government response indicators on exchange rate volatility are not affected by a change in the measurement of the COVID-19 pandemic.

This paper subsequently also carries out a robustness test of the various sub-indices under the closure and containment policy, economic response, and health system. The conclusions are still consistent with the basic results.[7]

(2) Robustness test: Using Sub-Samples

[7] Due to space limitation, the regression results of these two items are not displayed, but are available upon request to the author.

Table 6
Robustness test: Log(Bdeaths) and four comprehensive indicators.

Variable	(1)	(2)	(3)	(4)
L.Earn	0.045	0.055	0.252	0.108
	(0.697)	(0.153)	(0.998)	(0.763)
Log(Bdeaths)	1.851**	1.497**	0.833*	0.787
	(2.367)	(2.414)	(1.949)	(0.814)
ST	−0.114**			
	(−2.230)			
GR		−0.227**		
		(−2.485)		
CH			−0.153**	
			(−2.367)	
ES				−0.027
				(−0.438)
Constant	5.495	141.880*	92.907	177.215
	(0.069)	(1.844)	(1.007)	(1.251)
Observations	1,991	1,991	1,994	2,000
AR(1) (*p*-value)	0.001	0.074	0.002	0.002
AR(2) (*p*-value)	0.115	0.373	0.229	0.187
Sargan(*p*-value)	0.201	0.440	0.123	0.950

Notes: *z* statistics are in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. AR(1) and AR(2) denote the p-value for the Arellano–Bond test of first-order and second-order autocorrelations, respectively. Sargan represents the p-value of the over-identification test. In view of the limited space, the control variable, fixed effect, and other results are not reported, but are available upon request.

In view of the fact that China experienced a relatively serious pandemic in the early stage, and the government has adopted more stringent measures to almost shut down the Chinese economy, which is not experienced by other countries. Therefore, in order to ensure the robustness of the above conclusions, this study now excludes the Chinese samples and performs a sub-sample regression again. The regression results are still consistent with the above findings.

Table 7 provides the sub-sample regression results, which show that an increase in the percentage of biweekly confirmed cases significantly intensify exchange rate volatility, which is basically consistent with the above conclusion. Columns (1)–(4) respectively show the regression results of the government's four comprehensive indicators after excluding the Chinese sample. The regression results in Table 7 present that the stringency index, containment and health index, and economic support index all significantly negatively correlate at the 5% level, suggesting that government intervention in the prevention and control of the pandemic and economic support measures do help curb exchange rate volatility. Thus, the main findings continue to hold once the main explanatory variables are replaced and sub-sample regression is performed, credibly explaining the impact of COVID-19 and government intervention on exchange rate volatility.

5. Conclusion and policy implications

This research analyzes the relationship between COVID-19, government intervention measures, and exchange rate volatility. For empirical analysis, we use daily data of *ER*, measuring exchange rate volatility, biweekly confirmed COVID-19 cases, and data of government response indices from 20 countries over the period January 13 to July 21, 2020. We then apply the dynamic panel model with system GMM estimation to obtain the results.

Through theoretical and empirical analyses, this paper finds that an increase in the number of confirmed COVID-19 cases further intensifies exchange rate volatility. Regarding the regression results of the four comprehensive indicators in the government response system, the overall government response index, the containment and health index, the stringency index, and the economic support index all have a restraining effect on exchange rate volatility.

We next conduct regression analysis on various specific intervention measures under different categories and discuss the impact of different response policies on exchange rate volatility. For specific containment and closure categories, school closings, public transport closure, stay at home requirements, and restrictions on internal movement do play a role in restraining exchange rate volatility. On the other hand, the result of various measures under the economic response category indicates that income support policy, fiscal policy, and international aid also have a role in restraining exchange rate fluctuations. Finally, for the health system category, only public information campaign policy has a significant inhibitory effect on exchange rate volatility, whereas other measures are not statistically significant.

Based on the results of the theoretical and empirical analyses in this paper, a government's non-pharmaceutical interventions such as restrictions on internal movement and public information campaign can effectively reduce the uncertainty and panic caused by COVID-19, send positive signals to the markets and investors, and thereby restrain exchange rate volatility. Furthermore, economic support policies, such as income support and fiscal measures, may intensify private consumption, trigger changes in capital flows, and curb exchange rate volatility.

Table 7
Robustness test: Result by using sub-samples.

Variable	(1)	(2)	(3)	(4)
L.Earn	0.069*	0.012	0.038	0.016
	(1.672)	(0.158)	(0.634)	(0.300)
Log(Bcase)	0.860**	1.790*	1.696**	0.649**
	(2.331)	(1.656)	(2.067)	(2.526)
ST	−0.093**			
	(−2.290)			
GR		−0.216		
		(−1.645)		
CH			−0.205**	
			(−2.049)	
ES				−0.076**
				(−2.413)
Constant	−35.496***	−39.927**	−52.581***	−13.040
	(−3.061)	(−2.143)	(−2.647)	(−1.319)
Observations	1,894	1,894	1,894	1,894
AR(1) (*p*-value)	0.001	0.022	0.001	0.001
AR(2) (*p*-value)	0.305	0.143	0.154	0.948
Sargan(*p*-value)	0.257	0.165	0.695	0.123

Notes: *z* statistics are in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. AR(1) and AR(2) denote the p-value for the Arellano–Bond test of first-order and second-order autocorrelations, respectively. Sargan represents the p-value of the over-identification test. In view of the limited space, the control variable, fixed effect, and other results are not reported, but are available upon request.

Table A.1
OxCGRT indicators and component indicators.

Containment and closure	
C1	School closure
C2	Workplace closure
C3	Cancel public events
C4	Restrictions on gathering size
C5	Public transport closure
C6	Stay at home requirements
C7	Restrictions on internal movement
C8	Restrictions on international travel
Economic response	
E1	Income support
E2	Debt/contract relief for households
E3	Fiscal measures
E4	Giving international support
Health system	
H1	Public information campaign
H2	Testing policy
H3	Contact tracing
H4	Emergency investment in healthcare
H5	Investment in COVID-19 vaccines

Table A.2
Composition of comprehensive indicators.

Index	k	C1	C2	C3	C4	C5	C6	C7	C8	E1	E2	E3	E4	H1	H2	H3	H4	H5
Government response index	13	√	√	√	√	√	√	√	√	√	√			√	√	√		
Containment and health index	11	√	√	√	√	√	√	√	√					√	√	√		
Stringency index	9	√	√	√	√	√	√	√	√					√				
Economic support index	2									√	√							

Our research explores the relationships among COVID-19, government intervention, and exchange rate volatility for the first time in the literature. The results herein illustrate that the continued spread of COVID-19 does significantly raise exchange rate volatility. For government responses, the existing literature believes that intervention policies during the pandemic may have had a counterproductive effect on stock returns and caused poverty and inequality (Bonaccorsi et al., 2020; Tisdell, 2020). However, our research provides some different perspectives. Therefore, when a government adopts various intervention policies for COVID-19, it should fully consider the resultant impacts and try to eliminate any adverse ones on the overall economy.

Table A.3
The standard deviation of four currencies before and after COVID-19.

	2019	2020
USA/YEN	1.392	1.528
USA/RMB	0.077	0.064
GBP/USA	0.025	0.036
EUR/USA	0.009	0.024

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Tables A.1–A.3.

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